

PRACTICE WORKSHOP

Workshop 2: Control Flow, I/O and Exceptions

This workshop focuses on writing robust programs: looping constructs, reading files, and handling errors gracefully with try/catch instead of letting a program crash.

Learning Objectives

- Use for, while, and do-while loops appropriately.
- Read from and write to a simple text file.
- Catch and handle common runtime exceptions.
- Throw a custom exception for an invalid application state.

Exercises

Exercise 1: Number Guessing Game

Generate a random number between 1 and 100. Let the user keep guessing until they find it, printing 'Too high' or 'Too low' after each guess, and the number of attempts at the end.

Hints:

- `java.util.Random` or `Math.random()` can generate the number.
- Use a `while(true)` loop with a `break` condition.

Exercise 2: Safe Division

Write a method `divide(int a, int b)` that returns `a / b`. Wrap the call in a try/catch so that dividing by zero prints a friendly error message instead of crashing the program.

Hints:

- Catch `ArithmeticException` specifically before a general `Exception`.

```
public static int divide(int a, int b) {  
    // TODO: return a / b safely  
}
```

Exercise 3: Word Count from a File

Write a program that reads a text file line by line and prints the total number of words in the file. Handle the case where the file does not exist with a clear error message.

Hints:

- `BufferedReader` / `FileReader`, or `java.nio.file.Files.readAllLines`.
- Catch `IOException` / `FileNotFoundException`.

Stretch Goal (optional)

Create a custom exception class `InvalidAgeException` and throw it from a `setAge(int age)` method whenever a negative age is provided.

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